

The Evolution of AI

A timeline of changing approaches



AI

NAMASTE AI - SEASON 1

Episode 02

The Evolution of AI

Trace the lecture's journey from the question “Can machines think?” through rule-based systems, machine learning, deep learning, transformers, generative AI, ChatGPT, and today's agentic systems.

01:43:17

WebVTT transcript only

In this episode

01 What counts as artificial intelligence?**02** Turing, the imitation game, and the naming of AI**03** From rules to machine learning and deep learning**04** Computer vision and the language problem**05** Transformers, LLMs, and generative AI**06** The ChatGPT moment and agentic AI**07** Milestones, predictions, and the course ahead

Why begin with history?

00:00 – 05:02

The lecture opens with a promise: we will trace artificial intelligence from its earliest questions to the modern AI industry. The route includes rule-based AI, machine learning, deep learning, transformers, large language models, generative AI, and agentic AI.

Before going back in time, the instructor shows a Google list of the ten largest companies by market capitalization at the time of recording. He names NVIDIA, Apple, Alphabet, Microsoft, Amazon, Meta, Broadcom, Tesla, Taiwan Semiconductor, and Saudi Aramco. His observation is that nine of those ten companies are investing in technology and AI, while Saudi Aramco is the oil-company exception.

The wider point is not the exact ranking. Rankings change. The point is that money, research, and product development are flowing into AI. New models and tools are appearing across coding, images, video, and natural-language processing.

WHY THIS MATTERS

Today's AI products have a long background. Understanding that path creates a mental model: each new model is part of an evolution, not an isolated miracle.

The instructor asks learners to keep a notebook nearby and not skip the historical material. His teaching philosophy is simple: curiosity becomes stronger when we know the roots of a subject.

What is artificial intelligence?

05:02 – 08:04

The first concept arrives as a set of questions rather than a formal definition.

Q: If a machine can do these things, is it AI?

Play chess, detect spam, recommend movies, drive a car, write a poem, or generate a fictional selfie with a public figure?

A: In the lecture's framing, yes. Each task requires some form of intelligence.

Driving requires decisions. Recommending a movie requires some criteria. Writing a poem requires choices that not everyone can make easily. If a machine performs a task that normally requires human intelligence, the lecture places it under artificial intelligence.

The instructor does not claim to offer the only accepted definition. He gives a working definition that will guide the episode:

WORKING DEFINITION

Artificial intelligence is the science of making machines perform tasks that normally require human intelligence.

This definition is intentionally broad. It covers both older systems that made narrow decisions and modern systems that generate text, images, or actions.

“Can machines think?”

08:04 – 13:59

The lecture moves back roughly 75-80 years. Before 1950, machines were already valuable for fast calculations and defined mathematical operations. The change came when Alan Turing asked a different kind of question:

Can machines think?

The word **think** makes the problem difficult. How do we define thinking? How can we observe it? How can we decide whether a machine is intelligent without directly seeing an internal object called “thought”?

The lecture answers with the **Turing test**, based on an imitation game.

- 1 Place a human in one hidden room.
- 2 Place a machine in another hidden room.
- 3 A human judge asks both participants questions.
- 4 The judge reads the responses and tries to identify which participant is human.
- 5 If the judge cannot distinguish the machine from the human, the machine passes the test.

The important criterion is **indistinguishability through responses**. The test does not settle every philosophical question about thought, but it converts an abstract debate into something an observer can evaluate.

A field gets a name-and enters a cycle

14:01 – 18:59

The instructor credits computer scientist and mathematician **John McCarthy** with coining the term **artificial intelligence**. The main explanation says **1955**, while the later recap says **1956**.

Researchers gathered around an ambitious belief:

Every aspect of learning and intelligence could, in principle, be described precisely enough for a machine to simulate it.

The instructor emphasizes how bold this sounded before modern GPUs, internet-scale data, and today's AI software. It gave researchers a clear direction: if intelligence could be described precisely, perhaps a machine could reproduce it.

Progress did not follow a smooth upward line. AI went through repeated **hype cycles**:

1. Excitement and ambitious promises increased.
2. Results failed to match expectations.
3. Funding and attention fell.
4. A new approach renewed interest.

An **AI winter** is the low part of that cycle—a period with reduced hype, investment, research momentum, and visible breakthroughs.

Artificial intelligence or synthetic intelligence?

19:01 – 25:59

The lecture introduces a naming and philosophy debate from 1986. A researcher whose name is spelled in the transcript as **Haugeland** proposed the term **synthetic intelligence**.

The instructor first says “synthetic learning,” then immediately corrects himself to **synthetic intelligence**.

Why object to “artificial”? In ordinary speech, artificial hair or an artificial hand can mean something constructed or “fake.” Some researchers did not want machine intelligence to be treated as fake by definition.

Artificial intelligence

Presented here as simulating or imitating human intelligence. Decisions arise from training, algorithms, and programming. In this philosophical comparison, consciousness is not assumed.

Synthetic intelligence

Presented as genuine machine intelligence that learns and reasons independently, with the potential to become autonomous, self-directed, or aware.

This is the instructor's philosophical framing, not a complete standardized taxonomy. He connects it with a modern question: when an AI system writes code or speaks impressively, do users care whether its intelligence is “real” or “fake,” or do they care primarily about the result?

He argues that current systems sometimes perform better than an average person at particular tasks. That leads him to question an old assumption that machine capability would stop at the human level. The lecture does not settle consciousness or “real intelligence”; it uses the older debate to show that even the name of the field carried competing ideas.

Deep Blue: when intelligence became visible

26:00 – 29:59

In 1997, IBM's **Deep Blue** defeated **Garry Kasparov**, described in the lecture as the world's best or one of history's greatest chess players. Chess was associated with intelligence, so the public event produced a striking image: a machine had defeated an exceptional human in an intellectual game.

The instructor then adds an important qualification. Deep Blue, in his explanation, did not “think” in the human sense. It mathematically evaluated possible boards and moves. The exact mechanism is not developed in this episode, and the description is deliberately high-level.

Q: Was Deep Blue intelligent?

A: To observers, the win looked intelligent. The instructor nevertheless describes the system as move evaluation rather than genuine thought. The milestone matters because machine behavior became publicly competitive with human expertise.

This section also returns credit to early contributors. Contemporary conversations may focus on today's leaders and companies, but Turing and McCarthy helped create the questions and vocabulary that made the field possible.

Rule-based AI and expert systems

30:00 – 34:59

From the 1950s through the 1980s, much AI is described as **rule-based**. Imagine a very large chain of `if / else` decisions.

Example: a spam detector

```
if email contains "free"    -> possible spam
if email contains "$ $ $"  -> possible spam
if email contains "lottery" -> possible spam
```

Example: a medical expert system

```
fever + cold + body ache -> "It looks like flu."
```

The rules come from human experts. A doctor could write symptom rules; the machine would apply them. Such systems became known as **expert systems** because they tried to reproduce expert decisions.

WHAT WENT WRONG?

Attempt: Write enough rules to cover the world. **Problem:** variations are endless. A spammer can write `f-r-e-e`, add letters, or use punctuation. **Result:** manually enumerating every possibility does not scale.

That limitation creates the motivation for machine learning: instead of telling the machine every rule, give it examples and let it discover patterns.

Machine learning: learn from examples

35:00 – 38:03

The machine-learning philosophy in the lecture is:

Do not supply every rule. Supply labeled examples.

The central example is cat-versus-dog classification.

1 Collect a very large set of images.

2 Label each image `cat` or `dog`.

- 3 Train a model using those examples.
- 4 Give the trained model a new image.
- 5 Receive a prediction, possibly expressed as probabilities such as 80-90% for one class.

The transcript uses “one million” illustratively and alternates between one million total images and one million for each class. The exact count is not the teaching point; scale and labeling are.

This is a major change: the machine learns patterns from data. Yet people still help by defining useful features. A human might say that an elephant has a trunk, a camel has a hump, or a cat has whiskers. That manual feature work is one reason deep learning becomes important.

Deep learning: learn the features too

38:04 – 47:00

Researchers took inspiration from neurons in the human brain and built **artificial neural networks**. The important change in this lecture is automatic feature discovery.

In machine learning, a person may explicitly identify the eye, nose, or other features. In deep learning, the neural network can learn which visual patterns matter.

Examples named in the lecture include:

- finding a cat, person, mountain, or blue sky in a photo;
- face recognition and face unlock;
- speech recognition;
- translation;
- classifying unstructured inputs such as images, text, and audio.

Three conditions helped deep learning become practical:

More compute

The GPU revolution and faster chips made large neural-network training feasible.

More data

The growth of the internet and large datasets supplied many training examples.

Practical applications form a third force. Face unlock, image recognition, speech recognition, and translation made the technology useful, which attracted research and investment.

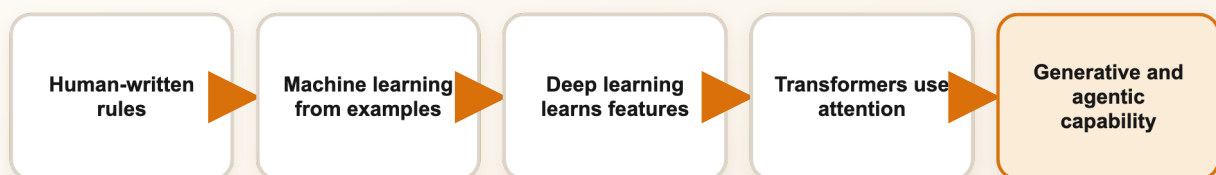
Machine learning and deep learning compared

Machine learning, as taught here	Deep learning, as taught here
Smaller datasets	Much larger datasets
Faster training	More compute-intensive training
Structured inputs	Can work with unstructured images, text, and audio
Human feature engineering	Automated feature learning
An ML algorithm uses extracted features	Neural networks learn internal features

The instructor deliberately does not teach neural-network mathematics, RNN training algorithms, or research details here. Those are complete subjects of their own.

From explicit rules to generative systems

The lecture presents these as major shifts in how capability is produced



The lecture's major shifts: hand-written rules give way to learned patterns, automatic feature learning, and transformer-based generation.

Computer vision: when machines could “see”

47:01 – 51:00

The lecture links the computer-vision leap to **ImageNet**, a large image dataset, and to **AlexNet** in 2012. The transcript is unsure about ImageNet's exact number of images/categories and badly transcribes the name of AlexNet's creator.

The stable concept is clear: training a deep neural network on a large, categorized image dataset greatly improved object recognition.

The applications named are broad:

- recognize people, animals, sky, trees, cars, and trucks in an image;
- support face unlock;
- provide visual input for self-driving cars;
- inspect X-rays for fractures, broken bones, or tumors;

- work with ultrasound reports;
- identify products in shopping applications.

The instructor summarizes the moment with one memorable phrase: **machines could see**. They could receive an image and identify objects or visual features inside it.

The medical examples are only application claims in the lecture. The episode does not discuss clinical accuracy, deployment boundaries, or diagnosis workflows.

Why language was so hard

51:01 – 56:00

Text can look easier than images, speech, or video, but the lecture says natural language became one of the hardest problems. The difficulty is ambiguity.

Q: “I saw a man with a telescope.” Who has the telescope?

A: The man may have it, or the speaker may be using it to see the man.

Q: “The chicken is ready to eat.” What is happening?

A: A cooked chicken may be ready to be eaten, or a living chicken may be ready to eat food.

The word **bank** creates another ambiguity:

- **river bank** refers to the land beside water;
- **Bank of India** refers to a financial institution.

The instructor playfully notes that a machine could misread **River Bank** as a financial bank named River, or **India** as a river name. Humans use context and experience to settle the intended meaning. Machines need a method for relating words across context.

The path toward longer context

56:01 – 01:01:00

The lecture sketches several earlier NLP approaches.

Bag of words

Classify a comment or paragraph by checking for words such as **good**, **impressive**, **amazing**, or **interesting**, versus **bad**, **unfit**, or **fake**.

The problem is that one word can mislead. **Terribly good** contains a negative-sounding word but expresses praise.

N-grams

Inspect two or three nearby words together. This adds local context, but a small group still cannot represent a full sentence or a distant relationship.

Recurrent neural networks (RNNs)

RNNs process longer sequences. The lecture gives a long sentence about a boy who wins a science competition after months of preparation, then uses the pronoun **he**. The system must link **he** back to **the boy**.

Long short-term memory (LSTM)

LSTM is introduced as an improved RNN with better memory and support for longer paragraphs/pages. Yet with a very long document such as a 100-page book, the system may lose early details by the time it reaches the end.

An Akshay-and-Aman example makes the same point: when a later sentence says **he scored very well**, the system must connect **he** with the correct person mentioned earlier.

Transformers: attention changes everything

01:01:01 – 01:05:59

The instructor treats the 2017 transformer architecture as the foundation of modern LLM products. He names the Google paper “**Attention Is All You Need**” and emphasizes **attention**—the ability to connect relevant pieces of context.

The beginner-level example is:

The lion did not cross the river because it cannot swim.

Here, **it** refers to **the lion**. The lecture uses this to illustrate that a model must relate words that are separated in the sentence.

SCOPE NOTE

This episode does not explain transformer internals. It marks the historical breakthrough and promises a later deep dive into attention and transformers.

The instructor calls transformers one of the most remarkable creations in human history. That is rhetorical emphasis from the lecture, not an objective ranking.

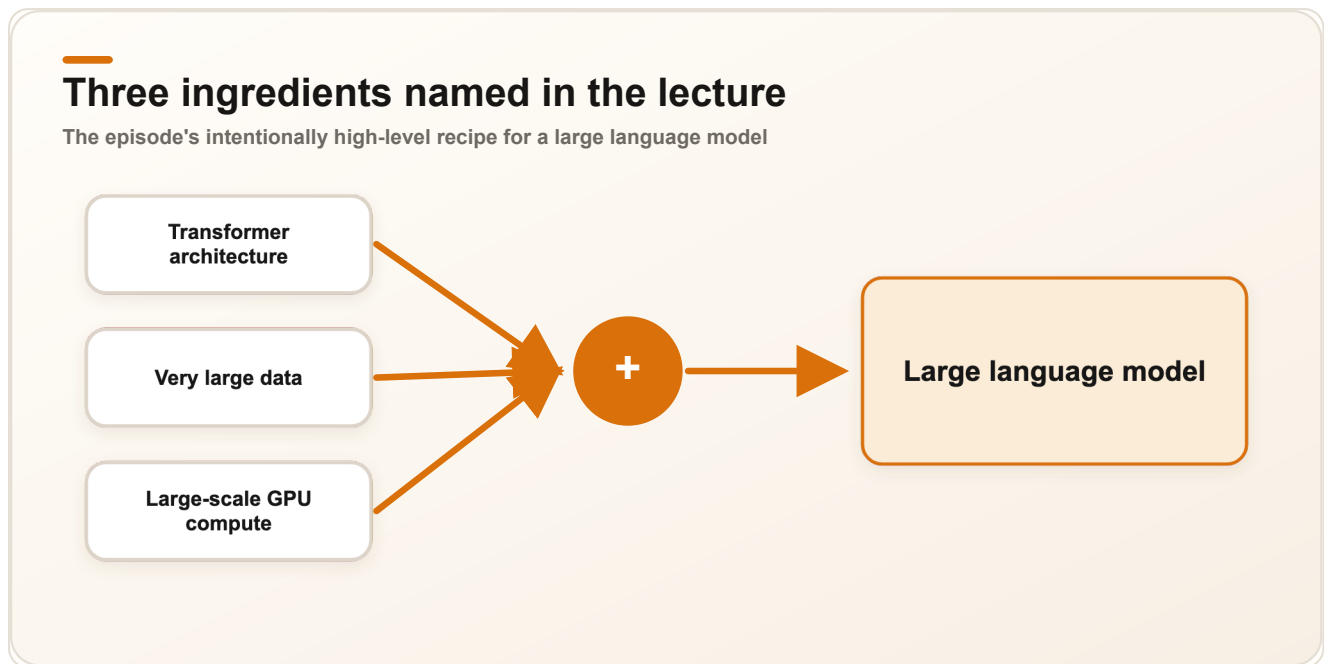
What makes a large language model possible?

01:06:01 – 01:08:59

At this episode's level, the instructor describes an LLM as:

A transformer trained on a very, very large amount of data.

Three ingredients come together:



The episode's intentionally high-level recipe for an LLM: transformer architecture, very large training data, and GPU compute.

- **Transformer architecture** provides the model structure.
- **Very large data** supplies patterns to learn.
- **GPU compute** supplies the processing capacity to train.

Training at that scale is expensive. The instructor uses this to explain why not every company or country creates a large model: the required money, data, and compute are not equally available. He names the United States, India, Bangladesh, Pakistan, Sri Lanka, and Nepal as examples in this discussion, along with Google, OpenAI, Anthropic, and Elon Musk's efforts.

The definition is intentionally incomplete and high-level. The lecture is establishing a historical relationship, not specifying a full model architecture.

Generative AI: from deciding to creating

01:09:01 – 01:12:59

The instructor draws a broad boundary between older and newer capability.

Earlier AI

Classify a cat or dog. Predict spam or not spam. Recommend a movie.

Generative AI

Create new text, poems, songs, jokes, images, videos, or documents.

The word to remember is **generate**. The model is not only choosing a label for an input; it produces a new output sequence.

Multimodal means working across multiple forms of content. The lecture names text, images, video, voice, and documents. A playful example is generating a fictional selfie with Ronaldo, Messi, Sachin Tendulkar, Dhoni, or another favorite public figure.

The instructor says this generation capability is where current investment is concentrated.

November 2022: the ChatGPT moment

01:13:00 – 01:18:00

The public release of **ChatGPT in November 2022** is presented as a turning point. Transformer research and LLM development already existed, but ChatGPT placed conversational generative AI directly in front of the public.

The teaching sequence is:

1. Transformers arrive in 2017.
2. Companies build large models behind the scenes.
3. OpenAI releases ChatGPT publicly in November 2022.
4. Users ask questions, request jokes, and hold conversations.
5. Adoption grows rapidly.
6. Competing products accelerate: Google/Gemini, X/Grok, Meta, Anthropic, and others.
7. The race expands across coding, images, video, mathematics, medicine, and other industries.

The lecture says ChatGPT can retain conversational context, such as remembering that a user earlier said they were a software engineer. It mentions **RLHF** but deliberately postpones the explanation.

The instructor also says companies trained models using scraped internet data “with permissions, without permissions.” The statement remains attributed to the lecture; no legal verification is supplied.

From a responder to an agent

01:18:00 – 01:21:00

Modern systems are described as multilingual and multimodal. The instructor names English, Hindi, Marathi, Gujarati, Konkani, Tamil, Arabic, French, Italian, and Chinese, as well as images, voice, video, and documents.

The larger change is from **answering** to **acting**. In the lecture's framing, present-day AI can:

- answer questions;
- think and plan;
- call APIs;
- write and debug code;
- use the web and tools such as calculators;
- create presentations and videos;
- remember context;
- make decisions and complete tasks;
- build, test, and deploy a program.

The instructor acknowledges the debate about whether AI will replace humans but does not resolve it. His direct claim is narrower: AI can already perform many tasks associated with human work, and its capability continues to improve.

The milestone timeline-and Move 37

01:21:00 – 01:26:59

The recap adds one major story that did not appear earlier: **AlphaGo**.

Google DeepMind's **AlphaGo** defeated Go champion **Lee Sedol** in 2016. The instructor describes Go as much more complex than chess and focuses on **Move 37** from the second match. According to the lecture:

- the move surprised Lee Sedol;
- it was not a move humans had previously played;
- it was recognized as a very strong move;
- it later became a known move among Go players.

These claims have not been externally verified in transcript-faithful mode.

The instructor asks interested learners to search YouTube for [Move 37 Lee Sedol vs AlphaGo match 2](#) and for the [AlphaGo documentary](#) / [AlphaGo – The Movie](#). No exact URL is supplied, so none should be invented.

Instructor's outlook: what may come next

01:27:00 – 01:34:00

This section is explicitly personal prediction, not guaranteed future behavior.

Personal and legacy agents

The instructor imagines an “Akshay Saini agent” that speaks in his style, as well as agents modeled on a grandmother or a deceased relative. With enough chat and communication data, future generations might interact with a digital persona after the original person has died.

Delegate routine work

Agents may handle tax filing, grocery orders, flight bookings, and email or WhatsApp replies. The lecture notes that Gemini can already draft email responses and imagines increasingly personalized or automatic replies.

Longer and more realistic generation

AI video may progress from clips of roughly 8-10 seconds to minutes or even full films. It may become difficult to distinguish generated video from recorded footage.

Multi-agent orchestration

Instead of one agent doing everything, several agents could behave like a company team:

- project-manager agent;
- developer agent;
- designer agent;
- tester agent;
- DevOps agent.

The developer agent could write code and pass it to the testing agent. Each agent specializes and hands work to the next.

Other future topics are named but not taught: reasoning models, RAG, MCP, robotics, and a Google item transcribed as `OKF / open knowledge format`. `[Transcript unclear]`

For robotics, the instructor searches for robots dancing at the Chinese Spring Festival Gala and uses their coordinated movement as an example of rapid improvement. No exact link is supplied.

What Namaste AI will-and will not-cover

01:34:00 – 01:43:17

The central closing idea is that today's AI was not created in a few recent years. Decades of questions, failed attempts, research, datasets, chips, and architectures came first. The rapid post-2017 period stands on that foundation.

The course will focus mainly on modern AI:

- how LLMs and ChatGPT work;
- transformers and attention;
- the “Attention Is All You Need” paper;
- agentic AI, planning, tool use, and API calls;
- relevant tools;
- practical projects and problem solving.

It will not deeply teach advanced mathematics, statistics, classical machine learning, or deep learning. Each is a large subject of its own. The series will bring in only the foundational pieces needed for the modern topics.

The instructor closes by encouraging notes, community participation, mutual help, public sharing, and feedback. He repeats that the history is not included merely for an interview question. It is there so learners can connect with the subject and stay curious.

Chapter summary

The episode begins with a broad definition: AI lets machines perform tasks that normally require human intelligence. It then follows the field from Turing's question and imitation game to the naming of AI, cycles of hype and winter, rule-based expert systems, data-driven machine learning, feature-learning neural networks, computer vision, and the struggle to understand language.

Transformers and attention become the decisive modern bridge. Combined with very large data and GPU compute, they make LLMs possible. Generative AI moves from classification and prediction to creating new outputs; ChatGPT turns that technical trajectory into a public conversational product; agentic AI extends responses into plans, tools, and tasks.

The final timeline and future outlook remind us to separate history, present capability, and prediction. The course will now concentrate on the post-2017 systems built on these foundations.

Key takeaways

- AI is framed as machines performing tasks that normally require human intelligence.
- The Turing test evaluates whether a judge can distinguish machine responses from human responses.
- Rule-based AI depended on expert-written decisions; machine learning learned from examples; deep learning also learned useful features.
- Language is difficult because meaning depends on context, not isolated words.
- Transformers, large data, and GPU compute are the lecture's three ingredients for modern LLMs.
- Generative AI creates; multimodal AI works across content types; agentic AI plans and acts.
- ChatGPT's public release in November 2022 is presented as the event that made conversational AI widely accessible.
- The future section is the instructor's outlook, not a set of verified guarantees.

Revision questions

1. How does the instructor define artificial intelligence?
2. Which six tasks does he use to test whether something counts as AI?
3. Why was “Can machines think?” different from asking machines to calculate faster?
4. How is the Turing test arranged, and what condition counts as passing?
5. What does the lecture mean by an AI winter?
6. How does the synthetic-intelligence debate differ from the usual artificial-intelligence framing?
7. Why did Deep Blue's victory look intelligent, and why does the instructor qualify that impression?
8. How does a rule-based spam detector work, and why does **f-r-e-e** expose a limitation?
9. What changes when a system learns from labeled examples?
10. What human assistance remains in the lecture's machine-learning explanation?
11. How does deep learning change feature extraction?
12. Which three conditions helped deep learning expand?
13. Why does the instructor describe computer vision as enabling machines to “see”?
14. Give both meanings of “I saw a man with a telescope.”
15. Why do bag of words and n-grams fail to capture full context?
16. What long-distance relationship is difficult in the RNN/LSTM examples?
17. What does attention connect in the lion-and-river sentence?

18. What three ingredients does the episode identify for an LLM?
19. How does generative AI differ from classification, prediction, and recommendation?
20. What does multimodal mean here?
21. Why is November 2022 treated as a turning point?
22. Which abilities make an AI system agentic in this lecture?
23. What was significant about AlphaGo's Move 37 according to the instructor?
24. Which future statements are observations, and which are predictions?
25. What topics will Namaste AI study deeply, and which subjects will remain high-level?